

JESS Thermodynamic Database v8.9

Chemical species in reactions with Iridium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
4MePyridine 4-Methylpyridine; Picoline (gamma); 4-Picoline				
0	108-89-4	68	C(6)H(7)N(1)	93.1283
Ba+2 Barium(II) ion				
2	22541-12-4	584	Ba(1)	137.330
Br2(l) Bromine, liquid				
0	---	285	Br(2)	159.808
Cd+2 Cadmium(II) ion				
2	22537-48-0	3554	Cd(1)	112.410
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl(1)	35.4530
Cl2(g) Chlorine gas				
0	7782-50-5	589	Cl(2)	70.9060
e-1 Electron				
-1	---	3399	E(1)	0.000000
Fe+2_Phenanth(3)				
2	---	12	C(36)H(24)Fe(1)N(6)	596.472
Fe+3_Phenanth(3)				
3	---	3	C(36)H(24)Fe(1)N(6)	596.472

H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H (1)	1.00794
H2O Water				
0	7732-18-5	5947	H (2) O (1)	18.0153
I2 (s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I (2)	253.800
Ir+1_4MePyridine_COD_Cl-1				
0	---	1	C (14) H (19) Cl (1) Ir (1) N (1)	428.985
Ir+1_4MePyridine (2)_COD_Cl-1				
0	---	1	C (20) H (26) Cl (1) Ir (1) N (2)	522.113
Ir+1_COD_Pyridine_Cl-1				
0	---	1	C (13) H (17) Cl (1) Ir (1) N (1)	414.958
Ir+1_COD_Pyridine (2)_Cl-1				
0	---	1	C (18) H (22) Cl (1) Ir (1) N (2)	494.059
Ir+1_I-1 (s) Iridium monoiodide				
0	---	1	I (1) Ir (1)	319.120
Ir+3 Iridium(III) ion				
3	22555-00-6	27	Ir (1)	192.220
Ir+3_Ba+2_Br-1 (6)				
-1	---	1	Ba (1) Br (6) Ir (1)	808.974
Ir+3_Ba+2_Cl-1 (6)				
-1	---	1	Ba (1) Cl (6) Ir (1)	542.268

Ir+3_Br-1(3)_(s) Iridium tribromide				
0	---	1	Br(3)Ir(1)	431.932
Ir+3_Br-1(6)				
-3	---	3	Br(6)Ir(1)	671.644
Ir+3_Cd+2_Br-1(6)				
-1	---	1	Br(6)Cd(1)Ir(1)	784.054
Ir+3_Cl-1(3)				
0	---	1	Cl(3)Ir(1)	298.579
Ir+3_Cl-1(3)_(s) Iridium trichloride				
0	---	1	Cl(3)Ir(1)	298.579
Ir+3_Cl-1(4)				
-1	---	1	Cl(4)Ir(1)	334.032
Ir+3_Cl-1(5)				
-2	---	3	Cl(5)Ir(1)	369.485
Ir+3_Cl-1(6)				
-3	---	8	Cl(6)Ir(1)	404.938
Ir+3_I-1(6)				
-3	---	1	I(6)Ir(1)	953.620
Ir+3_K+1_Cl-1(6)				
-2	---	1	Cl(6)Ir(1)K(1)	444.036
Ir+3_OH-1				
2	---	3	H(1)Ir(1)O(1)	209.227
Ir+3_OH-1(2)				
1	---	2	H(2)Ir(1)O(2)	226.235

Ir+3_OH-1(3)_ (s)				
0	---	2	H(3)Ir(1)O(3)	243.242
Ir+3(2)_S-2(3)_ (s) Diiridium trisulfide				
0	---	2	Ir(2)S(3)	480.620
Ir+4 Iridium(IV) ion				
4	22542-08-1	27	Ir(1)	192.220
Ir+4_Ba+2_Br-1(6)				
0	---	1	Ba(1)Br(6)Ir(1)	808.974
Ir+4_Br-1(6)				
-2	---	3	Br(6)Ir(1)	671.644
Ir+4_Cd+2_Br-1(6)				
0	---	1	Br(6)Cd(1)Ir(1)	784.054
Ir+4_Cl-1(2)_OH-1(4)				
-2	---	2	Cl(2)H(4)Ir(1)O(4)	331.155
Ir+4_Cl-1(3)				
1	---	1	Cl(3)Ir(1)	298.579
Ir+4_Cl-1(3)_OH-1(3)				
-2	---	3	Cl(3)H(3)Ir(1)O(3)	349.601
Ir+4_Cl-1(4)_OH-1(2)				
-2	---	3	Cl(4)H(2)Ir(1)O(2)	368.047
Ir+4_Cl-1(5)				
-1	---	7	Cl(5)Ir(1)	369.485
Ir+4_Cl-1(5)_OH-1				
-2	---	3	Cl(5)H(1)Ir(1)O(1)	386.492

Ir+4_Cl-1(6)				
-2	---	6	Cl(6)Ir(1)	404.938
Ir+4_I-1(6)				
-2	---	1	I(6)Ir(1)	953.620
Ir+4_Ru+4_Cl-1(10)_OH-1				
-3	---	1	Cl(10)H(1)Ir(1)O(1)Ru(1)	664.829
Ir+4_Ru+4_Cl-1(6)_OH-1(5)				
-3	---	1	Cl(6)H(5)Ir(1)O(5)Ru(1)	591.047
Ir+4_Ru+4_Cl-1(7)_OH-1(4)				
-3	---	1	Cl(7)H(4)Ir(1)O(4)Ru(1)	609.492
Ir+4_Ru+4_Cl-1(8)_OH-1(3)				
-3	---	1	Cl(8)H(3)Ir(1)O(3)Ru(1)	627.938
Ir+4_Ru+4_Cl-1(9)_OH-1(2)				
-3	---	1	Cl(9)H(2)Ir(1)O(2)Ru(1)	646.384
Ir+4_S-2(2)_(s) Iridium disulfide				
0	---	2	Ir(1)S(2)	256.340
Ir(s) Iridium				
0	7439-88-5	16	Ir(1)	192.220
Ir2O3(s) Diiridium trioxide				
0	1312-46-5	2	Ir(2)O(3)	432.438
IrI2(s) Iridium diiodide				
0	---	1	I(2)Ir(1)	446.020
IrO(s)				

0	---	1	Ir(1)O(1)	208.219
IrO2 (s) Iridium oxide; Iridium dioxide				
0	12030-49-8	2	Ir(1)O(2)	224.219
IrO4-2				
-2	---	1	Ir(1)O(4)	256.218
K+1 Potassium(I) ion				
1	24203-36-9	462	K(1)	39.0980
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Pyridine Pyridine; Azine; py				
0	110-86-1	297	C(5)H(5)N(1)	79.1014
Ru+4 _Cl-1 _OH-1(5)				
-2	---	1	Cl(1)H(5)O(5)Ru(1)	221.562
Ru+4 _Cl-1(2) _OH-1(4)				
-2	---	1	Cl(2)H(4)O(4)Ru(1)	240.007
Ru+4 _Cl-1(3) _OH-1(3)				
-2	---	1	Cl(3)H(3)O(3)Ru(1)	258.453
Ru+4 _Cl-1(4) _OH-1(2)				
-2	---	4	Cl(4)H(2)O(2)Ru(1)	276.899
Ru+4 _Cl-1(5) _OH-1				
-2	---	3	Cl(5)H(1)O(1)Ru(1)	295.344

S-2 Sulfide ion; Sulphide ion				
-2	14127-58-3	466	S(1)	32.0600
S(s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S(1)	32.0600